

YASHWANTHSAI PATHIPATI

Software Engineer | C/C++ Systems & Distributed Infrastructure

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PROFESSIONAL SUMMARY

Software Engineer with 1.5+ years of production C/C++ experience building high-performance distributed systems and database infrastructure at Zoho Corporation. Designed and shipped a PostgreSQL extension for horizontal sharding across worker nodes, with deep focus in memory management, query optimization, multithreading, and low-level performance profiling. Completed M.S. in Computer Science (GPA 3.97) at the University of South Florida.

EDUCATION

Master of Science in Computer Science

August 2024 – May 2026

University of South Florida, Tampa, Florida | GPA: 3.97/4.00

Bachelors in Electronics and Communications Engineering

August 2019 – May 2023

Sri Venkateswara University College of Engineering, Tirupati, India | GPA: 8.34/10.00

EXPERIENCE

Member Technical Staff

February 2023 – July 2024

Zoho Corporation Private Limited | Chennai, India

- Engineered and maintained production C/C++ backend systems serving concurrent multi-tenant workloads, improving reliability and reducing system downtime by ~30% through targeted memory management fixes and runtime profiling.
- Designed and implemented a production PostgreSQL extension in C for hash-based horizontal sharding across 4+ worker nodes, enabling parallel data processing and reducing single-node query load by ~60% under concurrent workloads.
- Reduced query latency by ~35% by profiling execution paths with GDB and Valgrind, identifying and resolving memory leaks and suboptimal index traversal patterns in the PostgreSQL query planner.
- Improved cluster throughput by ~25% by redesigning shard-key routing and rebalancing data distribution across nodes, reducing peak CPU load under high-concurrency workloads from ~85% to ~63%.
- Owned end-to-end SDLC for 5+ features from requirements and system design through implementation, unit testing, deployment, and production support by independently driving each to release.
- Led zero-downtime data migrations and system upgrades for a production cluster, coordinating rollout across multiple nodes with rollback plans achieving 100% uptime across all deployment events.
- Contributed to PostgreSQL logical replication pipeline by implementing mutex-based concurrency controls to safely coordinate WAL (Write-Ahead Log) stream access across parallel replication workers, eliminating race conditions under concurrent subscriber loads.
- Reduced post-release defect rate by ~20% by enforcing modular C/C++ design patterns, introducing unit test coverage for critical paths, and conducting structured code reviews across the team.
- Collaborated with cross-functional teams of 8+ engineers in an Agile environment to resolve critical stability issues, cutting mean time to resolution (MTTR) by ~40% through systematic root-cause analysis and shared debugging sessions.

Web Developer Intern

May 2022 – August 2022

ZettaByte SoftTech | Tirupati, India

- Built and integrated 5+ RESTful APIs in Node.js, reducing frontend data-fetch latency by ~20% and enabling seamless communication between React UI and backend services.
- Improved application responsiveness by ~25% through performance-focused JavaScript optimizations, including lazy loading and reducing unnecessary re-renders.
- Participated in Agile sprints across a 3-month cycle, contributing to sprint planning, debugging, and integration testing.

- Delivered 3 production-ready features meeting all client requirements and passing QA with no defects reported post-launch.

TECHNICAL SKILLS

Languages	C, C++ (STL, RAII, templates), Python, Java
Concurrency & Systems	Distributed Systems, Multithreading, Concurrency, Mutex, Lock-free, Memory Management, Custom Allocators, RAII, Performance Tuning, System Design
Databases	PostgreSQL, MySQL, Query Optimization, Data Partitioning
Backend	REST APIs, Node.js, Flask
High-Perf Computing	OpenCL, CUDA (basic), GPU Optimization
Tools	Git, GitHub, Linux (Ubuntu), Bash, GDB, Valgrind, CMake, Docker
AI/ML	PyTorch, Transformers, Knowledge Distillation, LoRA

PROJECTS

C++ Memory Management Framework — *C++, Systems Programming, Custom Allocator* | github.com/PathipatiYashwanthSai/Memory-Management-Framework

Designed and implemented a hierarchical memory context system in C++ inspired by PostgreSQL's memory management architecture, enabling structured allocation, context switching, and recursive context-tree teardown. Each allocation stores ownership metadata with 8-byte aligned payloads; released chunks are retained in a free list for O(1) reuse, avoiding repeated heap calls. Implemented typed object lifecycle management via C++ templates supporting RAII-compliant construction and destruction. Demonstrates custom allocators, pointer arithmetic, metadata-based memory tracking, free-list reuse, and memory safety patterns applicable to high-performance systems and database engine internals.

Mini Distributed PostgreSQL Extension — *C, PostgreSQL, Distributed Systems* | github.com/PathipatiYashwanthSai/mini-distributed-postgres

Built a C-based PostgreSQL extension that distributes table data across multiple worker nodes using hash-based sharding, implementing coordinator metadata tables for worker registration, distributed table tracking, deterministic shard-to-worker mapping, and shard-key based query routing. Supported distributed insert and point-lookup operations across Dockerized PostgreSQL worker nodes, with benchmark scripts to measure throughput, latency, and scalability under concurrent workloads.

Sentiment Analysis via Knowledge Distillation & LoRA — *Knowledge Distillation + LoRA* | github.com/PathipatiYashwanthSai/Sentiment-Analysis-Using-Knowledge-Distillation

Applied Knowledge Distillation and LoRA fine-tuning to compress BERT into a DistilBERT student model that is 46% faster and half the memory size, while achieving 90% accuracy and 91% F1 on SST-2 within 1% of the teacher. On an unseen IMDB benchmark, the distilled model outperformed BERT (88.72% vs 86.08% accuracy), demonstrating robust generalization under compression.

DSP Learning Platform — *React, Flask, Signal Processing* | github.com/PathipatiYashwanthSai/DSP-Algorithms

Led a 4-member team to design and build an interactive DSP learning platform, implementing FFT, convolution, and digital filtering algorithms to process user-defined signal inputs with real-time outputs. Structured the codebase with strict separation between the computation engine and UI layer, reducing debugging time across a 4-month development cycle.